Sampling & Estimation

Point estimates for the population mean

1 A sample of 6 measurements has values $14, 18, 12, 16, 15, 17$. Find the unbiased point estimate of the population mean.

2 Explain why the unbiased estimator of population variance, $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$, divides by $n-1$ rather than n.

3 A sample of $n = 8$ has $\sum x_i = 152$. Find $\bar{x}$, the unbiased point estimate of μ.

Computing the unbiased sample variance

4 For the sample $5, 8, 10, 7, 10$, find the unbiased estimate of the population variance.

5 A sample of $n = 10$ has $\sum(x_i - \bar{x})^2 = 81$. Find the unbiased estimate of the population variance and standard deviation, to three decimal places.

6 For the sample $12, 9, 14, 11, 13, 15$, find the unbiased estimate of the population standard deviation, to three decimal places.

The Central Limit Theorem

7 State the Central Limit Theorem as it applies to the sample mean $\bar{X}$ of a random sample of size n from a population with mean μ and variance σ^2.

8 A population has mean $\mu = 40$ and standard deviation $\sigma = 12$. A random sample of size $n = 36$ is taken. Using the Central Limit Theorem, state the approximate distribution of $\bar{X}$.

9 Explain why the Central Limit Theorem is so useful in practice, even when the underlying population distribution is unknown or non-normal.

Using the distribution of the sample mean

10 Using $\bar{X} \sim N(40, 4)$ from above, find $P(\bar{X} > 43)$, given $\Phi(1.5) \approx 0.9332$.

11 Using the same distribution, find $P(38 < \bar{X} < 41)$, given $\Phi(0.5) \approx 0.6915$ and $\Phi(-1) \approx 0.1587$.

Confidence intervals for μ with known σ

12 A sample of $n = 49$ has mean $\bar{x} = 68$ and known population standard deviation $\sigma = 7$. Construct a 95% confidence interval for μ, using $z^* = 1.96$.

13 A sample of $n = 64$ has mean $\bar{x} = 25$ and $\sigma = 8$. Construct a 90% confidence interval for μ, using $z^* = 1.645$.

Confidence intervals with unknown σ (large samples)

14 A sample of $n = 100$ has mean $\bar{x} = 52.4$ and sample standard deviation $s = 9$. Since n is large, construct a 99% confidence interval for μ using $z^* = 2.576$ as an approximation, to two decimal places.

15 A sample of $n = 50$ has mean $\bar{x} = 110$ and sample standard deviation $s = 15$. Construct a 90% confidence interval for μ, using $z^* = 1.645$, to two decimal places.

Determining the required sample size

16 A researcher wants a 95% confidence interval for μ with margin of error at most 1.5, given $\sigma = 10$. Find the minimum sample size, using $z^* = 1.96$, rounding up.

17 A researcher wants a 99% confidence interval for μ with margin of error at most 3, given $\sigma = 15$. Find the minimum sample size, using $z^* = 2.576$, rounding up.

Interpreting confidence intervals

18 A 95% confidence interval for a population mean is $(41.2, 48.8)$. State the correct interpretation of this interval.

19 Explain why increasing the confidence level from 90% to 99%, with the sample data unchanged, produces a wider confidence interval.

A well-designed word problem: estimating average commute time

20 This question has four parts. A researcher surveys $n = 64$ randomly selected commuters, finding sample mean commute time $\bar{x} = 34.5$ minutes with sample standard deviation $s = 12$ minutes. (a) State the unbiased point estimate of the population mean commute time. (b) Since n is large, construct a 95% confidence interval for the true mean, using $z^* = 1.96$. (c) The city claims the average commute is 30 minutes. Using your confidence interval, comment on whether this claim is plausible. (d) The researcher wants a margin of error of at most 2 minutes at the same confidence level. Find the minimum required sample size, using $s = 12$ as an estimate of σ, rounding up.